# Machine Learning Methods for Predicting March Madness Tournament Outcomes

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CS 536 Final Project

## I. Introduction

March Madness is one of the most difficult sports tournaments to predict because of its single-elimination format. Unlike a long regular season, where stronger teams have many games to prove themselves, one poor tournament performance can end a team's season. This makes bracket prediction a challenging problem for machine learning.

The purpose of this project is to build a machine learning model that predicts outcomes in the men's and women's NCAA basketball tournaments. The project uses historical NCAA data including regular season results, tournament results, tournament seeds, team information, and ranking data. Instead of predicting an entire bracket in one step, we treat each tournament game as an individual matchup prediction problem.

The project focuses mainly on three goals. First, we preprocess the data so that it can be used by machine learning models. Second, we develop a model that predicts game outcomes and can be used to fill out a bracket. Third, we explore second-chance brackets, where predictions are updated once the tournament field has been reduced from 64 teams to 16 teams.

## II. Research Questions

Here are a list of questions which sparked our interest for this project and guided our experiments:

1) Can regular season team statistics be transformed into matchup-level features that meaningfully predict NCAA tournament outcomes?
2) Do ranking-based features such as seed, NET-style rankings, and PageRank improve prediction compared to using raw basketball statistics alone?
3) Can an XGBoost regression model predict tournament point differential well enough to produce calibrated win probabilities?
4) Do the important predictive features differ between the men's and women's tournaments?
5) Does prediction become more difficult in second-chance brackets beginning at the Sweet 16, where remaining teams are more similar in strength?

## III. Assumptions and Simplifications

Several assumptions were made in order to make the problem manageable. First, we represent team strength mostly through season averages. This simplifies the modeling problem, but it also hides game-to-game variation. A team that improved late in the season may look weaker than it actually is, while a team that declined late may still look strong based on its full-season averages.

Second, our PageRank method assumes that losing by a larger margin transfers more ranking power to the winning team. This helps the model account for strength of victory, but it can also overemphasize blowouts or games where the final score was affected by garbage-time minutes.

Third, the XGBoost model assumes that patterns learned from past tournaments are useful for future tournaments. To reduce overfitting, we use season-based validation, where one season is held out at a time. This better matches the real prediction task because a bracket model must predict a future tournament using past seasons.

Finally, we simplify bracket prediction by treating each game mostly as an independent matchup. In reality, tournament games are connected because injuries, fatigue, travel, confidence, and previous tournament performance can affect later rounds.

## IV. Data

### A. Raw Data

The first goal of the project is to convert the raw NCAA datasets into a structured format that can be used for modeling. The raw data is separated into multiple CSV files, such as regular season results, tournament results, seeds, teams, conferences, and ranking metrics. Since a machine learning model needs one organized table of inputs and outputs, the first major step is to clean, combine, and reshape these datasets. We obtain the datasets from the Kaggle March Machine Learning Madness Website [1].

### B. Preprocessing

**Dual Perspectives**

The most important preprocessing step is converting the original winner-loser format into a matchup format. In the raw detailed results files, games are stored with columns such as:

`WTeamID, WScore, LTeamID, LScore`

This means that the winning team is always listed first. If this format were used directly, the model could accidentally learn that the first team always wins, instead of learning from the actual statistics.

To fix this, our code creates two versions of each game. In the first version, the winning team is listed as Team 1 and the losing team is listed as Team 2. In the second version, the losing team is listed as Team 1 and the winning team is listed as Team 2. This creates a balanced matchup structure.

**Target Metric**

For every matchup, the target variable is point differential:

$$\text{PointDiff} = \text{Team}_1\text{Score} - \text{Team}_2\text{Score}$$

If PointDiff $> 0$, then Team 1 won. If PointDiff $< 0$, then Team 1 lost. This lets the model learn from both perspectives of the same game, and as a bonus, doubles the total amount of training and testing data available. The key reason we choose Point Differential instead of a binary win/loss metric is that it helps our model understand relative strength differences of varying magnitudes between teams.

**Regular Season Feature Creation**

After the games are reshaped, the next step is to create regular season team statistics. For each team in each season, the code groups regular season games by team and calculates average statistics. These averages summarize how a team performed before the NCAA tournament.

Examples of regular season features include:

- average score,
- average field goals made and attempted,
- average three-pointers made and attempted,
- average free throws made and attempted,
- average offensive and defensive rebounds,
- average assists,
- average turnovers,
- average steals,
- average blocks,
- average personal fouls,
- average point differential,
- average opponent score,
- average opponent field goals made and attempted.

These regular season averages are then merged into the tournament matchup table. As a result, each tournament game contains Team 1's regular season statistics and Team 2's regular season statistics.

The general structure of the final modeling table is:

$$\text{Team}_1\text{ID}, x_{1,1}, x_{1,2}, \ldots, x_{1,k}, \text{Team}_2\text{ID}, x_{2,1}, x_{2,2}, \ldots, x_{2,k}$$

where $x_{1,k}$ is the $k$-th feature for Team 1 and $x_{2,k}$ is the same feature for Team 2.

We also introduce an additional k columns that list the difference in each corresponding feature between the teams by subtracting Team 2's value in that statistic from Team 1's value in that statistic. This gives us the ability to pass in features that more directly represent the gap between both teams in a matchup. This helps us see whether a statistic is actually useful for prediction. For example, if teams with better average point differential tend to win tournament games, then average point differential is likely a useful feature. This connects to the "Moneyball" idea of identifying which statistics actually matter for team success.

We also compute a few additional features derived from other columns in the table such as:

- average three-point field goal percentage
- average field-goal percentage
- average free-throw percentage
- average assist-to-turnover ratio

The reasoning behind these features is that they are directly reflective of a team's offensive efficiency, something which we hypothesized would have a strong correlation with winning.

Some features provided in the dataset that we ignored were:

- Home vs. Away Game: tournament games are typically at a neutral venue. The rare case we can see this being applicable is if the model is 50-50 for a particular matchup, but a tournament game is happening very close to one of the team's home base, local fans rallying them could provide some sort of homecourt advantage. Also, traveling to play in a significantly different timezone can be disadvantageous. There is evidence suggesting this for NFL teams [8].
- Conference: Although it is true that certain conferences have dominated basketball in waves, and that set of conferences is rather small, we believe that the other data does a better job of reflecting current team strength than affiliation with strength in the past.
- Coaching: Once again, this is one of those features that could potentially be relevant for 50-50 games, but we did not incorporate it because we estimate its overall importance to be low.

One important consideration here is that strength of schedule varies greatly across teams in NCAA college basketball, and the raw statistics themselves are not necessarily the best indicators of relative team strength by themselves. This highlights the importance of unifying team strength across teams that don't frequently play each other, which leads us to our next section: Seed and Ranking Features.

**Seed Features**

Tournament seed is also included because it is one of the strongest baseline indicators of team strength. In the raw seed data, seeds appear as strings such as `W01`, `X04`, or `Y12`. The code extracts the numeric part of the seed and converts it into an integer.

For each matchup, the model uses:

$$\text{T1\_seed}$$

$$\text{T2\_seed}$$

$$\text{seed\_Diff} = \text{T2\_seed} - \text{T1\_seed}$$

A positive seed_Diff means that Team 1 has the better seed. For example, if Team 1 is a 2-seed and Team 2 is a 10-seed, then:

$$\text{seed\_Diff} = 10 - 2 = 8$$

This gives the model a simple way to understand seed advantage.

We also examined and considered many other ranking systems for addition to the model. Some of the key ones include BPI, KenPom Rankings, KPI, NET Ranking, and Wins Above Bubble (WAB), all of which are considered by the NCAA selection committee themselves when making the seeding [2]. The committee also uses other criteria besides raw ranking data such as region to determine the structure of the bracket.

We believe it is valuable to understand the key metrics and input that goes into some of these ranking systems, so some of the most popular ranking systems are described below. Note that our experiments used more than just these 4 systems and Figures 2 through Figure 8 will show a variety of systems.

*BPI*:

- Coach's past performance
- Recruiting ranking
- Current roster returning
- Performance of returning players

*KenPom*:

- Effective field goal percentage
- Turnover percentage
- Offensive rebounding percentage
- Free throw rate

*NET Ranking*:

- Adjusted net efficiency rating
- Team Value Index (rewards teams for road wins and defeating quality opponents)

*WAB Ranking*:

- How much better you did than if a standard team (NET Ranking = 45) was placed in your exact circumstances

**PageRank**

One of our biggest contributions here was adapting the PageRank [4] methodology (used by Google for its search engine to rank webpages) towards ranking basketball teams.

The key idea here is to consider each basketball team as a node in a directed graph. We consider each team to have some sort of power, and that power flows to it from the teams that it defeated and flows away from it to the teams that it lost to.

Specifically, focusing on teams that it lost to, we first do a bit of preprocessing where if a team lost to another team multiple times, we standardize by saying that it lost to that team by the average of the differentials in its losses.

Then, we add up the total margin of defeat across all teams, and divide every loss by that amount to normalize the outgoing power flow to be 1.

This becomes a column in the PageRank matrix, where each column represents a team's loss fingerprint, and the nonzero rows represents teams that it lost to and how badly.

For example, if Team A loses to Team B by an average of 8 points and loses to Team C by an average of 12 points, then the total loss differential is:

$$8 + 12 = 20$$

The normalized edge from Team A to Team B is:

$$\frac{8}{20} = 0.40$$

The normalized edge from Team A to Team C is:

$$\frac{12}{20} = 0.60$$

This means more ranking power flows to the team that beat Team A by more points.

Essentially, the key idea of PageRank is that we can assess the strength of teams in the following fashion:

1) Create a directed graph in the manner described above.
2) Randomly choose a node in that graph.
3) Move along a directed edge with probability = the weight of that edge (more likely we move to a team that defeated us by a larger margin).
4) Repeat this process, and record how much time we spend at each node.
5) Every so often, randomly jump to another spot in the graph in order to ensure coverage of disconnected components.

The PageRank equation used is:

$$v = dMv + \frac{1-d}{N}\mathbf{1}$$

where $v$ is the PageRank vector, $d$ is the damping factor, $M$ is the adjacency matrix, $N$ is the number of teams, and $\mathbf{1}$ is a vector of ones. This is often known as the Power Method for computing PageRank.

The damping factor controls how strongly the ranking follows the graph. A larger $d$ gives more importance to following power flow in a section. A smaller $d$ makes power flow mechanism more often randomly jump around to cover various parts of the graph. In our testing, we experimented with different damping and tolerance values to see which setting produced the strongest relationship with tournament point differential, and settled at a damping factor of 0.1.

The PageRank algorithm is applied season by season. For each season, the code produces a PageRank power score for every team:

PR_Power

These scores are merged into the tournament matchup table as:

T1_PageRank

T2_PageRank

PageRank_Diff

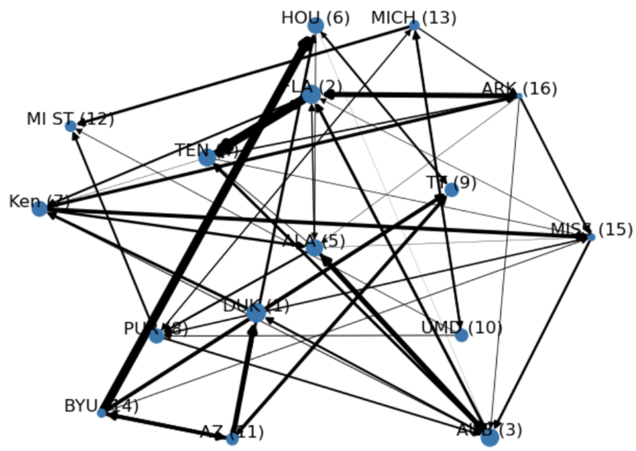


Fig. 1. PageRank Graph Results from 2025 Sweet 16. We see that Duke and Florida, the title favorites, emerged as the top teams. PageRank value is represented proportionally to node size. If teams played each other during the season, edge thickness represents the strength of the power transfer between them.

## Correlation Exploration - Ranking Systems

Before we dive into the model creation and feature selection, we first wanted to examine some data trends to get a better feel for which features really matter. In order to do this, we plotted Point Differential vs. every feature differential, and also computed correlation coefficients, or $R^2$ values. The data used for this was every tournament game.

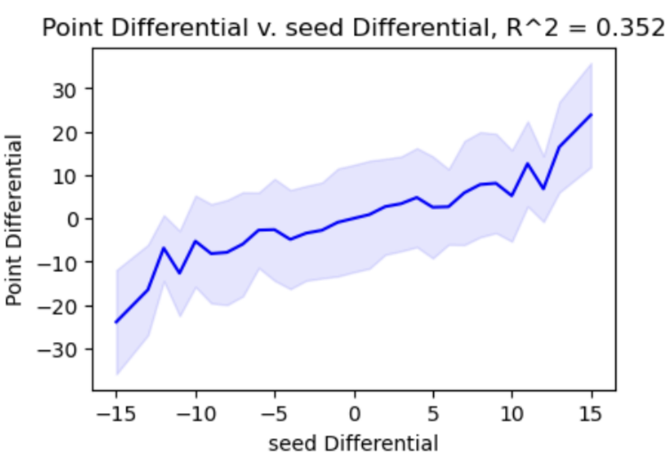


Fig. 2. Plot of Tournament Game Point Differential vs. Seed Differential

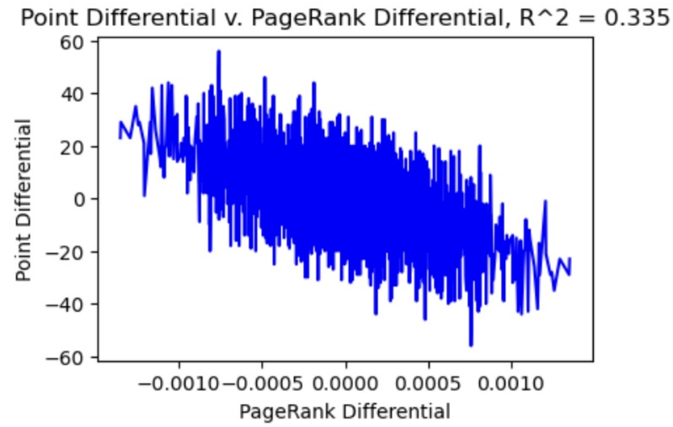


Fig. 3. Plot of Tournament Game Point Differential vs. PageRank Differential

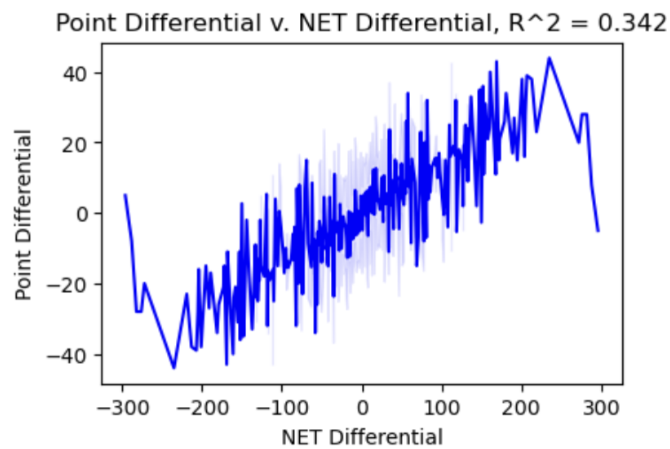


Fig. 4. Plot of Tournament Game Point Differential vs. NET Ranking Differential

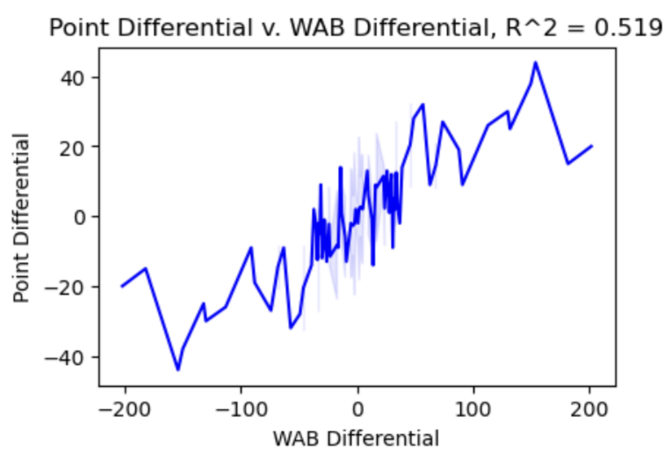


Fig. 5. Plot of Tournament Game Point Differential vs. WAB Ranking Differential

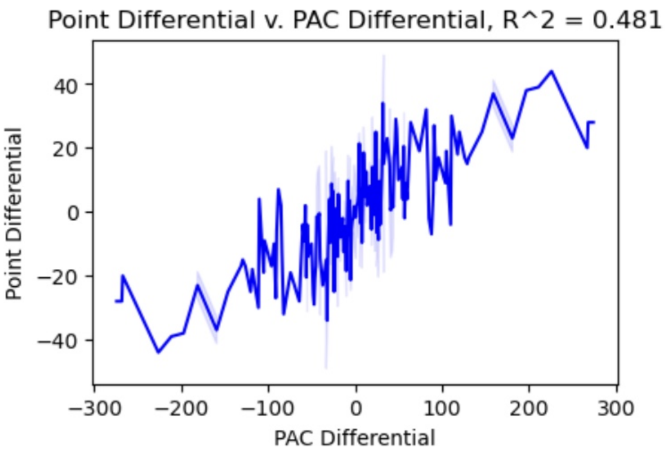


Fig. 6. Plot of Tournament Game Point Differential vs. PAC Ranking Differential

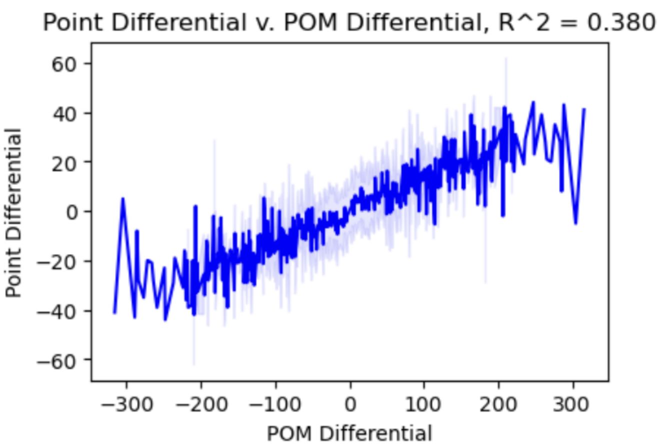


Fig. 7. Plot of Tournament Game Point Differential vs. POM Ranking Differential

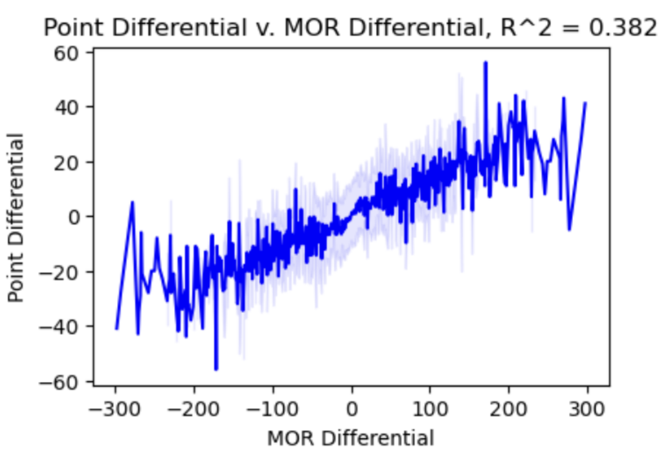


Fig. 8. Plot of Tournament Game Point Differential vs. MOR Ranking Differential

**Correlation Exploration - Statistics**

When examining which gameplay statistics are most correlated with point differential, we see that the difference between average point differential between 2 teams, average field goals made, average assists, and average score are some of the more correlated features. However, the correlation signals are not very strong or robust as shown by the large amount of noise in the plots.

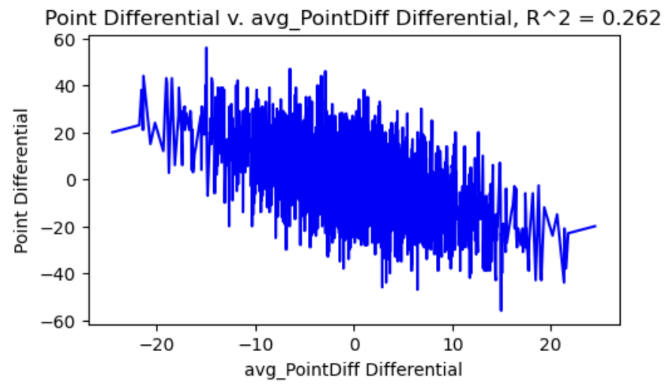


Fig. 9. Plot of Tournament Game Point Differential vs. Average Regular Season Point Differential

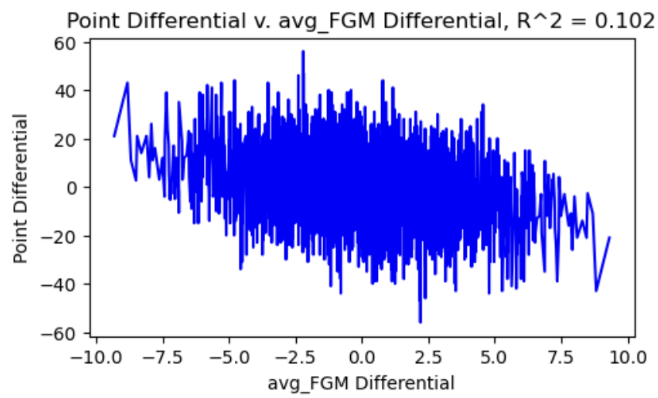


Fig. 10. Plot of Tournament Game Point Differential vs. Average Regular Season Field Goals Made Differential

| **Statistic** | $\mathbf{R^2}$ |
|---|---|
| Average Score Differential | 0.090 |
| Average FG Attempted Differential | 0.046 |
| Average 3's Made Differential | 0.011 |
| Avg. Offensive Rebounds Diff. | 0.041 |
| Avg. Defensive Rebounds Diff. | 0.031 |
| Avg. Assists Diff. | 0.078 |
| Avg. Turnover Diff. | 0.034 |
| Avg. Block Diff. | 0.058 |

Table of $R^2$ values for various statistic differentials

**Correlation over Time**

We also plot correlation of the magnitude of the feature differentials over time to see if there have been any changes in what is vital for a winning recipe.

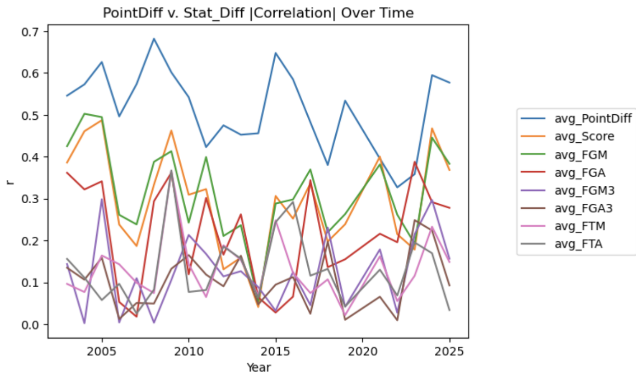


Fig. 11. Correlation Between Point Differential and Scoring Statistic Differential Magnitude Over Time

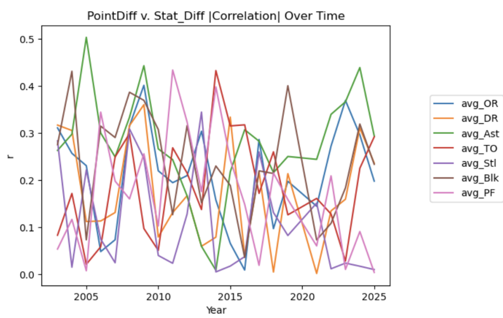


Fig. 12. Correlation Between Point Differential and Other Statistic Differential Magnitude Over Time

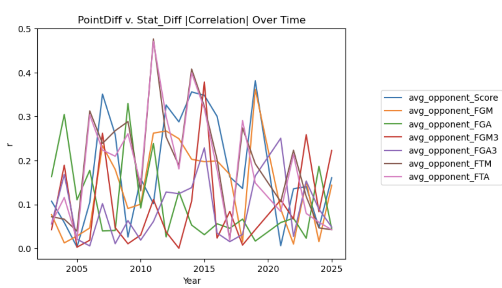


Fig. 13. Correlation Between Point Differential and Opponent Scoring Statistic Differential Magnitude Over Time

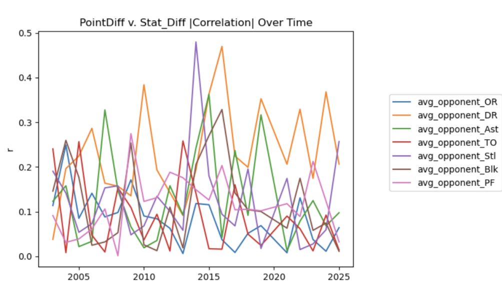


Fig. 14. Correlation Between Point Differential and Other Opponent Statistic Differential Magnitude Over Time

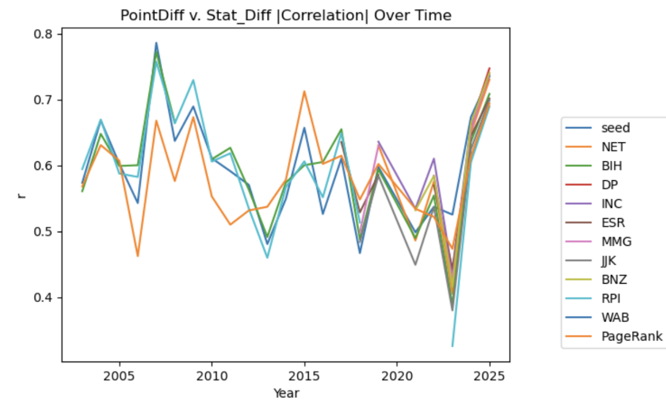


Fig. 15. Correlation Between Point Differential and Ranking Differential Over Time

### Discussion

Based on the graphs above, there are no clear trends in a certain statistic correlating more with point differential over time. This is contrary to our expectations because there has been a lot of talk about the rise in importance of 3 point shooting, so we expected to see positive trends in corresponding statistics. This suggests that models derived from data from the other years should be mostly applicable when testing on a holdout year.

In general we notice that metrics that are very close to capturing regular season point differential and scoring are the most strongly correlated with point differential during the tournament itself, suggesting the model's natural bias towards offense-related features since there is a more direct connection there.

Another note is that the ranking system trends move together for the most part, indicating that a lot of the ranking systems do a good job of encoding relative team strength, and that there could be a strong correlation between the underlying statistics and parameters used to do so.

### Notes about Women's and Second Chance Brackets

Some existing work in this domain merged women's and men's data into one dataframe, and included a binary parameter (0 for men, 1 for women). This approach gives a larger amount of data for the model to train on, and is convenient since it doesn't involve separate preprocessing, but we felt that it was flawed in the sense that it may not fully discern differences in signal importance for both the men's and women's tournament. Empirically, upsets have been significantly less common in the women's bracket, so we didn't want to simply hope that the model picks up on these via the gender binary variable, but ensure that the models are tuned appropriately via separation of data and training processes.

Another goal of the project is to explore second-chance brackets. A second-chance bracket begins after some tournament games have already been played. For example, after the first two rounds, the field is reduced from 64 teams to 16 teams. At that point, the model can update predictions using the teams that are still alive.

Second-chance brackets are useful because many original brackets are damaged by early upsets. Once a major team is eliminated, the original bracket may no longer be realistic. A second-chance model gives us a chance to predict the rest of the tournament using updated information.

This task is also harder in some ways. By the Round of 16, the remaining teams are usually stronger and closer in quality. There are fewer weak teams left, so the model cannot rely as much on obvious seed mismatches. This means features such as recent tournament performance, point differential, and matchup strength may become more important.

For the second-chance bracket setting, the model can be adjusted to focus only on the teams that remain in the tournament. Instead of starting from the full field, the model begins from the Round of 16. The code can rebuild the bracket using only these teams and then predict the remaining rounds.

For the second chance data, we moved the first two rounds (Round of 64 and Round of 32) from the test set to the training set. We wanted to make use of all information available at the time of making Second Chance Brackets.

## V. MODEL

The second goal is to develop a model that can predict tournament games and evaluate its effectiveness. The project uses a matchup-by-matchup prediction method. This means the model predicts the outcome of individual matchups as a predicted Point Differential. Then we map that to a win probability between 0.01 and 0.99, using a function that fits point differential to win percentage on training data. For example, if we predict a point differential for Team 1 to beat Team 2 by $\approx 10$ points, and 80% of the times where we predict a team to win by $\approx 10$ points it actually wins, then we assign Team 1 a 0.8 win probability. An example of such a function fit to data is provided below.

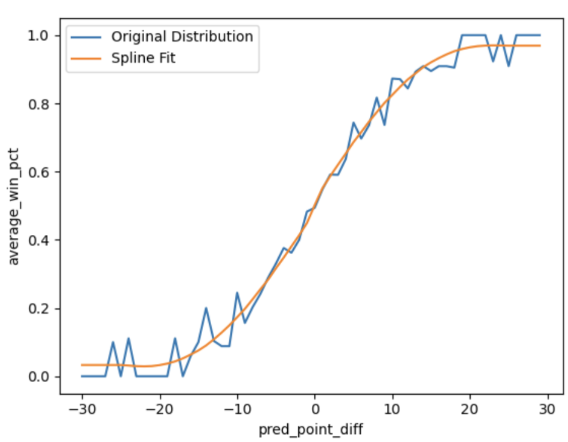


Fig. 16. Line is fit to existing data results. That line will be used for mapping from prediction point differential space to prediction probability space.

### Brier Score Metric

We use the Brier score to evaluate the quality of our predicted win probabilities. This is a standard used in Kaggle's March Madness Competitions. It is defined as follows:

$$\text{Brier Score} = \frac{1}{n}\sum_{i=1}^{n}(f_i - o_i)^2$$

where:

- $f_i$ : probability that an event was forecast
- $o_i$ : actual outcome of the event (0 or 1)

This metric rewards the model for being more confident in its predictions, but it also doesn't overly penalize it for being wrong. This is fair because of the inherent heap of variance in March Madness.

### XGBoost Model

The model used in this project is XGBoost. XGBoost is a gradient boosted decision tree model. It builds many small decision trees, where each new tree tries to correct errors from the previous trees. This makes it useful for tabular sports data because it can model nonlinear relationships between features. Another reason we chose this model is because the Kaggle Tournament has shown that time and time again, most of the best models either entirely rely on XGBoost, or use an ensemble of models where XGBoost plays a key role.

For example, a team's scoring average may be important, but its effect may depend on other features such as turnovers, shooting efficiency, defensive strength, or seed. XGBoost can capture these kinds of interactions better than a simple linear model.

In our code, the model is trained using the processed tournament matchup table. The feature vector for one game is:

$$X = [x_{1,1}, \ldots, x_{1,k}, x_{2,1}, \ldots, x_{2,k}, x_{D,1}, \ldots, x_{D,l}]$$

where the first group of features belongs to Team 1 and the second group belongs to Team 2. The last group is the subset of the k features that we computed a differential of between the teams.

The model can be trained in two possible ways. One option is classification, where the model directly predicts whether Team 1 wins:

$$y = \begin{cases} 1, & \text{if Team 1 wins} \\ 0, & \text{if Team 1 loses} \end{cases}$$

However, in our implementation, we mainly use XGBoost regression to predict point differential:

$$y = \text{PointDiff}$$

This is useful because point differential gives more information than a win/loss label. If the model predicts Team 1 by 12 points, that is a stronger prediction than if it predicts

Team 1 by 1 point. The predicted point differential can later be converted into a win probability.

**Training and Validation Method**

To evaluate the model, the code uses a season-based validation approach. Instead of randomly splitting games, the model holds out one season at a time. For example, the model can train on all seasons except 2023 and then test on the 2023 tournament.

This is important because March Madness prediction is a future-season problem. In a real bracket prediction setting, we would train on past seasons and predict a new tournament. Holding out full seasons better matches that real-world situation.

For each held-out season, the code:

1) selects all other seasons as training data,
2) selects the held-out season as testing data,
3) trains the XGBoost model,
4) predicts point differential,
5) evaluates prediction error.

One metric used for point differential prediction is mean absolute error:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

where $y_i$ is the actual point differential and $\hat{y}_i$ is the predicted point differential. This is not used as a training loss function, but rather just a metric to help understand how well the model is doing in a way that is easy to contextualize, i.e. this model's predictions, are on average, off by 10 points. The training loss function used in the decision tree is Mean Squared Error Loss, which is differentiable.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

## VI. Experiments and Results

We tried out many different combinations of features, and also experimented with some intermediate steps and hyperparameters with the goal of finding optimal settings for model performance.

| **Statistic Set, $S_i$** | $\mathbf{|S_i|}$ | **MAE** | **Brier** | **Tree Depth** |
|---|---|---|---|---|
| $S_1$ | 30 | 8.928 | 0.1858 | 4 |
| $S_2$ | 14 | 9.011 | 0.1880 | 4 |
| $S_3$ | 72 | 8.987 | 0.1876 | 4 |
| $S_4$ | 15 | 9.045 | 0.1881 | 4 |
| $S_5$ | 37 | 8.883 | 0.1855 | 4 |
| $S_{all\_diff}$ | 67 | 8.921 | 0.1865 | 4 |
| $S_{seed\_diff}$ | 1 | 9.163 | 0.1920 | 4 |
| $S_{all}$ | 201 | 8.946 | 0.1862 | 5 |
| $S_{all}$ | 201 | 8.907 | 0.1854 | 4 |
| $S_{non\_rank\_diffs}$ | 29 | 9.836 | 0.2065 | 4 |
| $S_{offense\_diffs}$ | 12 | 10.328 | 0.2173 | 4 |
| $S_{defense\_diffs}$ | 13 | 11.023 | 0.2338 | 4 |
| $S_{5\_norm\_no\_round}$ | 37 | - | 0.1855 | 4 |
| $S_{all\_norm\_no\_round}$ | 37 | - | 0.1854 | 4 |

- $S_1$: seed, PageRank, couple ranking systems, all associated ranking diffs, T1 and T2 score, blocks, fouls, field goals made & attempted, PointDiff (self and opponent averages)
- $S_2$: seedDiff, few ranking system Diffs, PointDiff_Diff, Assist to Turnover Ratio Diff, Free Throws Made Diff, Free Throw Percentage Diff, Offensive & Defensive Rebounding Diff, Steal, Block, Turnover Diffs
- $S_3$: T1 and T2 almost all stats, only Diffs used for ranking systems
- $S_4$: T1 and T2 seed, score, blocks, personal fouls, field goals attempted, personal fouls, opponent field goals attempted
- $S_5$: seeds, ranks, self and opponent score, blocks, personal fouls, free throw percentage, offensive rebounding, defensive rebounding, three point percentage

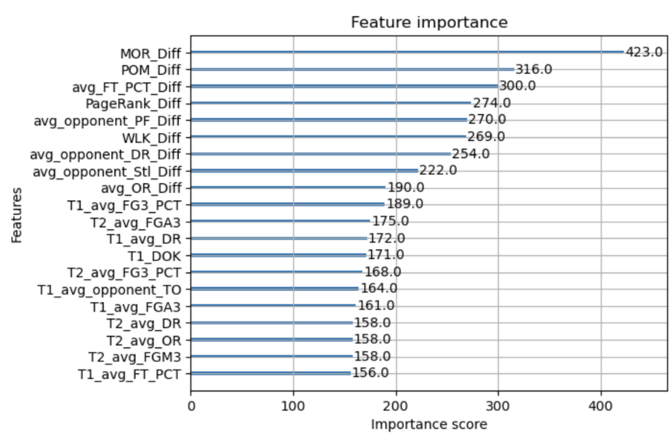


Fig. 17. Feature Importance map for Model trained on all but 2025 data

**Analysis of Results - Men's 64 Team Bracket**

Based on our results for the Men's Tourney for all 64 games, we can draw a few conclusions from our data.

1) **Optimal Number of Features**: While it is true that our best Brier score was achieved by using all 201

features, including aggregate team stats, team differentials, dozens of ranking metrics, and all aggregate opponent statistics, a lot of these statistics encode very similar information, so they are somewhat redundant. It seems that a reasonable number of features is in the 25-40 range because we can achieve low Brier Scores and MAE for our predictions without relying on too much data or too much data with overlapping underlying information. Compared to the baseline of one feature used of seed differential, the model can pick up more details about the underlying distribution with more features.

2) **Rankings are Important Unifying Metric**: Albeit all the madness that comes and the thrill of higher seeds getting bounced by Cinderella teams, seed and rankings still tend to be stable predictors for team success on average. We see this in our correlations, and our feature importance chart, where differences in ranking systems appear at the top. Also, in our trials without using seed as a unifying metric for team strength, Brier Score and MAE are considerably worse
3) **Offensive Features were More Predictive than Defensive Features**: Based on our tests, neither offense nor defense alone can predict too well without unifying ranking metrics, though offense can predict better than defense. There remains some doubt associated with this because offensive results are inherently easier to record and more correlated with score, which may minimize the ability for a model to actually gauge defensive effectiveness fairly.
4) **Free Throws are a Key to Success**: Across a lot of our models, average free throw percentage differential consistently came up as a top metric for the decision trees, indicating that it is an important factor in the fine margins for March Madness. Interestingly, the earlier correlation charts did not show that free throw percentage had a strong correlation with point differential. One reason this may be the case is how feature importance is measured by decision trees, in this case gain. The tree may have found this to be a useful feature when determining games, even though it may be drowned out by noise in the aggregate data set.

**Analysis of Results - Women's 64 Team Bracket**

For the women's tournament, we had higher MAE values, ranging from 10-11 but lower brier scores, falling in the 0.140 to 0.150 range. We applied similar testing procedures for parameters, and optimal results came when restricting the set of features to 15-20, where 5 or so were ranking based, and the rest were derived from season aggregates involving point differential, blocks, and personal fouls. The baseline model only using seed differential achieved a Brier score of 0.1500 and an MAE of 11.261, performing worse than models that combined features. One of the best performing models had a Brier score of 0.1445 and an MAE of 10.306

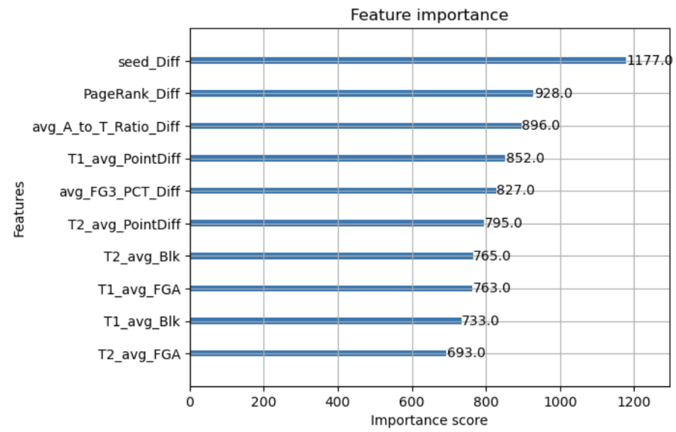


Fig. 18. Feature Importance Map for 2025 Women's Tournament

As expected, seed differential and PageRank differential were the most important features used in the XGBoost decision trees. Historically, better seeds are more consistently better in the women's game than in the men's game. We have included notebooks containing the same detailed graphical analysis for the Men's section in the code files for this project. What stood out to us is the importance of 3 point shooting and assist to turnover ratio. These are more important in the women's tournament than in the men's tournament, indicating differences in styles of basketball, perhaps tied to physicality.

**Analysis of Results - Men's Sweet 16 Bracket**

For the men's sweet 16 onwards predictions, naturally we see Brier score and MAE metric performance decline since later rounds inherently have stronger teams that are closer in skill. The best results (Brier score: 0.224, MAE: 9.12) are achieved by models containing 25-30 features, some ranking features, along with core defensive and offensive efficiency metrics. The feature importance plot for this, along with the other statistic correlation plots are provided in the code files. Seed no longer plays as important of a role in the decision tree in terms of importance, but free throw shooting percentage dominates across many trials. The number of personal fouls committed and drawn also appear as important variables at this later stage in the tournament. Ranking systems such as RPI emerge as strong predictors of success. RPI factors in a team's winning percentage, along with opponents' winning percentage, and opponents' opponents' winning percentage to unify team strength. PageRank also appears consistently as a Top 10 important metric in the decision tree.

**Analysis of Results - Women's Sweet 16 Bracket**

For the women's sweet 16 onwards predictions, here we also see Brier score and MAE metric performance decline since later rounds inherently have stronger teams that are closer in skill. The best results (Brier score: 0.176, MAE: 10.42) are achieved by models containing 25-30 features, some ranking features, along with core defensive and offensive efficiency metrics. The feature importance plot for this, along with the other statistic correlation plots are provided in the code files. Difference in rebounding, point differential, three point shooting percentage, and PageRank emerge as the most important features here.

## VII. Takeaways

This project shows that March Madness prediction benefits from combining several types of information rather than relying on a single feature group. Seed and ranking systems were consistently useful because they provide a broad estimate of team strength across different schedules. Regular season box score statistics added more detail, but many individual statistics were noisy when viewed on their own. This suggests that raw statistics are useful, but they need to be combined with ranking-based features that account for opponent quality.

The PageRank method was useful because it provided a graph-based way to measure team strength. Instead of only counting wins or losses, PageRank allowed strength to flow through the network of teams based on regular season results. This helped address the fact that NCAA teams do not all play each other and often have very different strengths of schedule. However, the PageRank method also depends heavily on how edges are defined and weighted. Our choice to weight losses by average margin of defeat is reasonable, but it may overemphasize blowouts or games that were not representative of a team's true strength.

The XGBoost model was effective because it could combine seed, ranking, statistical, and PageRank features in one tabular model. It also allowed nonlinear relationships between features, which is important because basketball outcomes are rarely explained by one statistic alone. However, the results also show the risk of using too many features. The all-feature models performed well, but smaller feature sets often achieved similar results, suggesting that many features contained redundant information.

The second-chance bracket experiments showed that prediction becomes harder in later rounds. This makes sense because the Sweet 16 contains stronger teams that are closer in ability. As a result, the model cannot rely as much on obvious seed mismatches, and smaller factors such as free throw percentage, rebounding, fouls, and PageRank become more important.

Overall, the project suggests that machine learning can improve matchup-level prediction, but it cannot remove the uncertainty that makes March Madness difficult. The most responsible interpretation of the model is probabilistic: it can identify stronger and weaker matchups, but it should not be treated as a guaranteed bracket generator.

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